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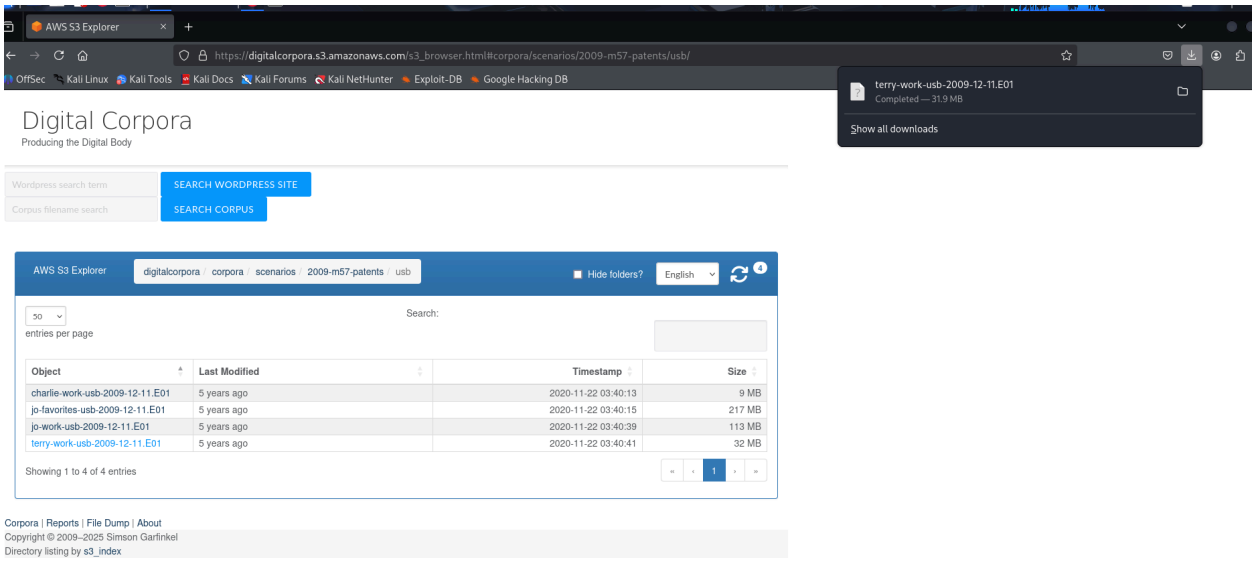
Autopsy Analysis of M57 USB Image

In this lab I used Autopsy to analyze a forensic image from the M57 USB collection. The goal was to load the image, set up a new case, explore the file system, review metadata, check deleted files, and document the key findings. I captured screenshots at each step to show the exact analysis process.

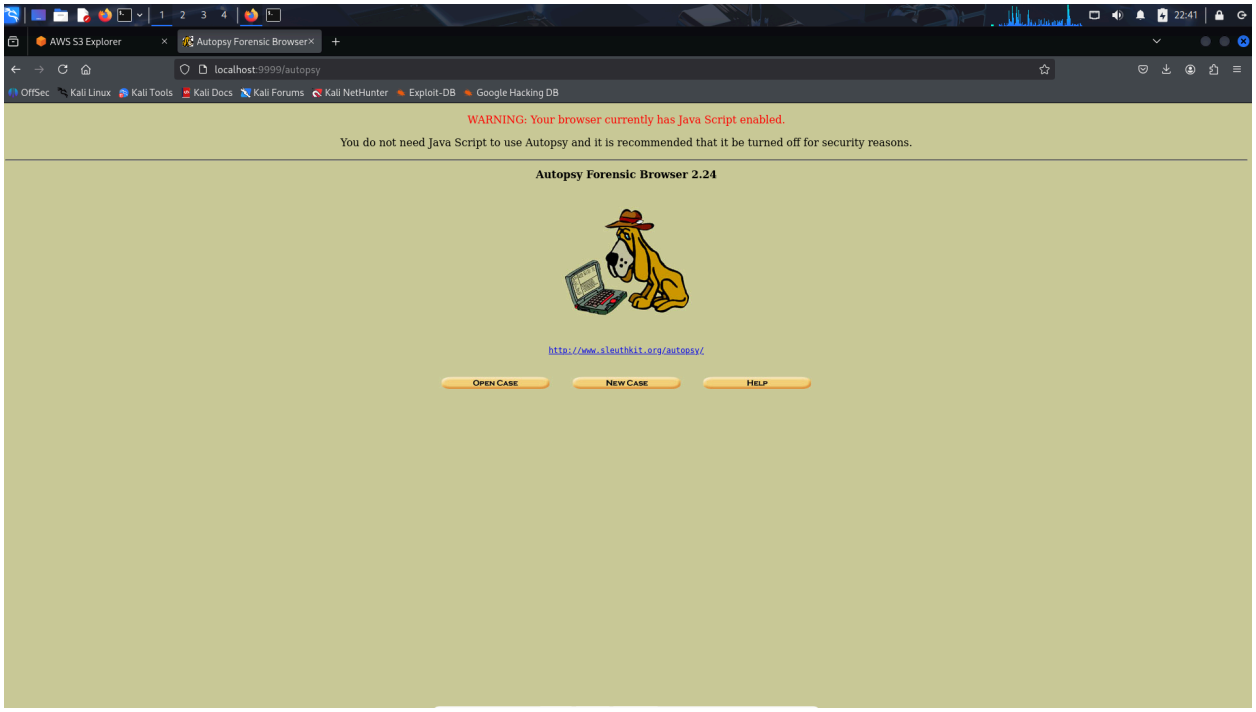
First I made the work folder for storing images. Then I launched Autopsy with the sudo autopsy command. Autopsy started on port 9999 and displayed the evidence locker location.



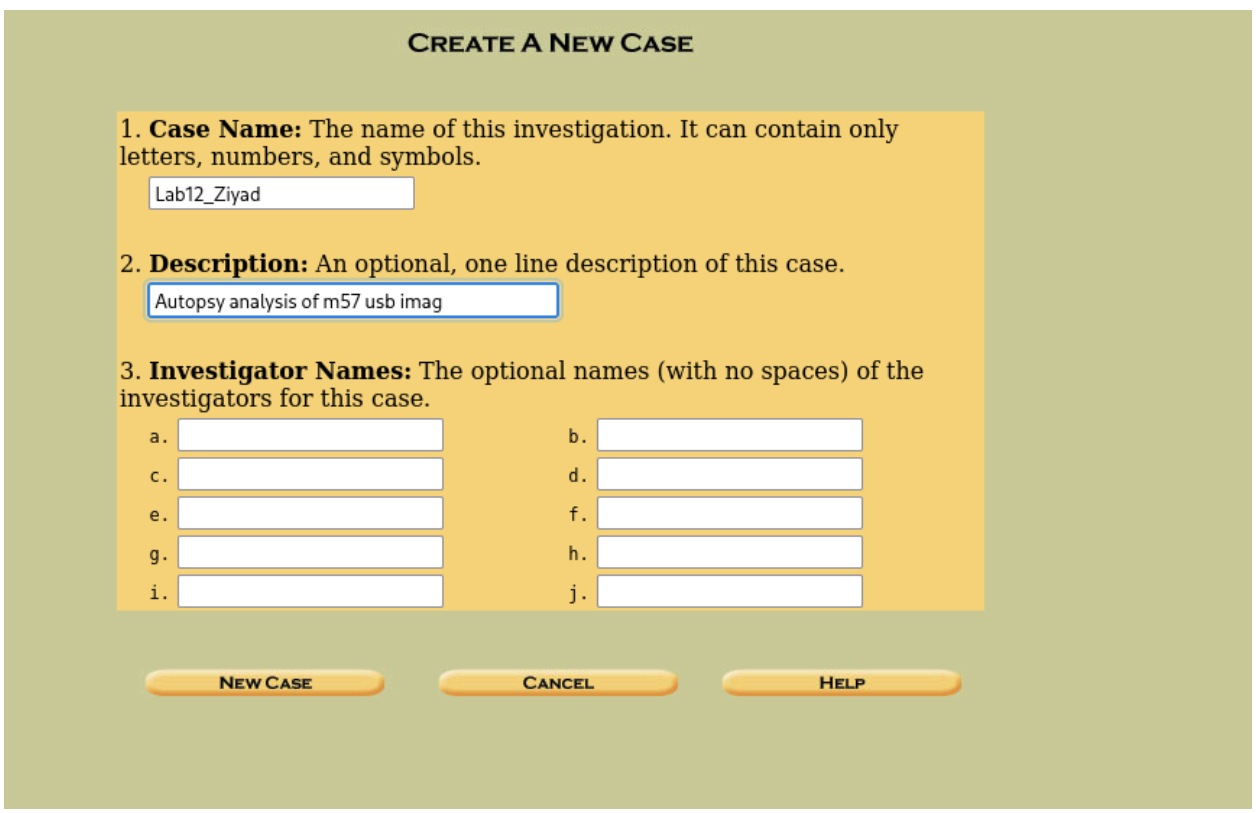
I opened a browser and went to localhost on port 9999 where the Autopsy interface loaded.

```
File  Actions  Edit  View  Help
┌──(ziyad㉿kali)-[~]
└─$ mkdir -p ~/work/images
┌──(ziyad㉿kali)-[~]
└─$ sudo autopsy
[sudo] password for ziyad:
═══════════════════════════════════════════════════════════════════════
                    Autopsy Forensic Browser
                http://www.sleuthkit.org/autopsy/
                          ver 2.24
═══════════════════════════════════════════════════════════════════════
Evidence Locker: /var/lib/autopsy
Start Time: Fri Nov 14 22:39:31 2025
Remote Host: localhost
Local Port: 9999

Open an HTML browser on the remote host and paste this URL in it:
    http://localhost:9999/autopsy

Keep this process running and use <ctrl-c> to exit
```

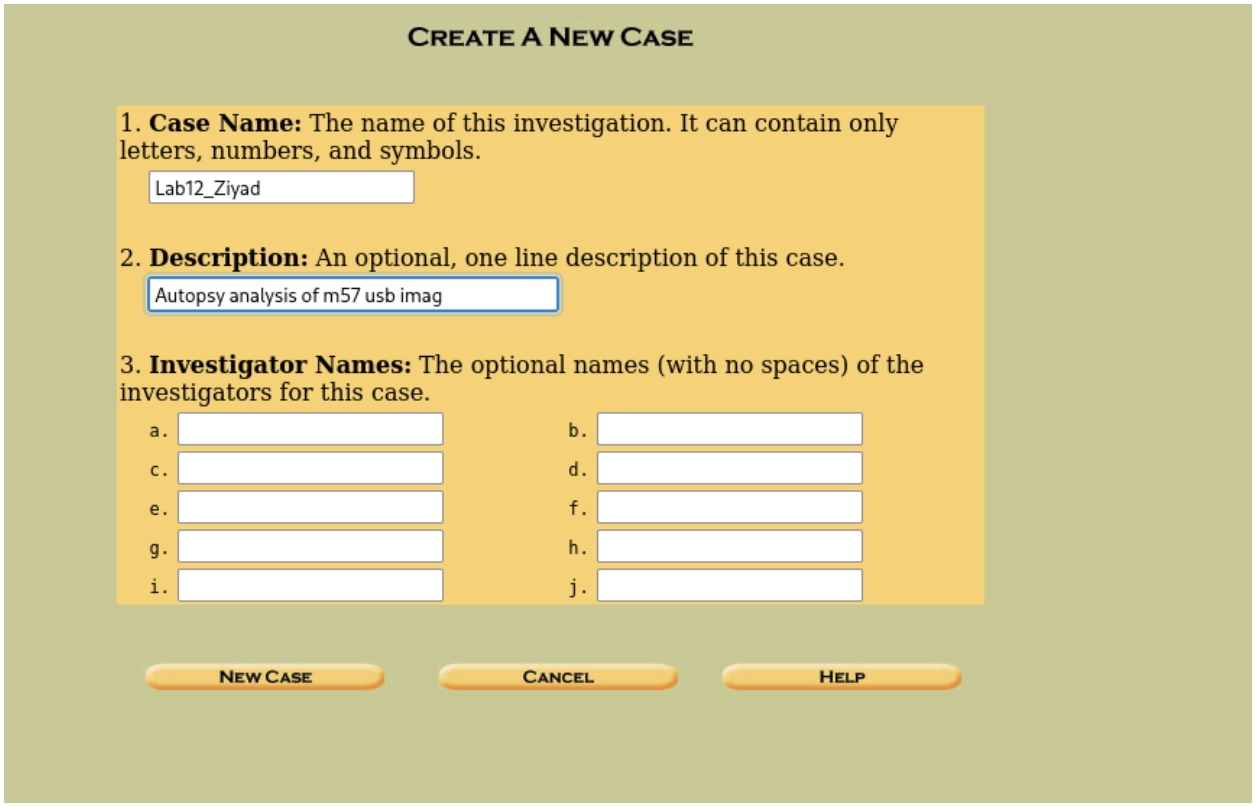
I created a new case named Lab12 Ziyad and added a short description explaining the purpose of the case.



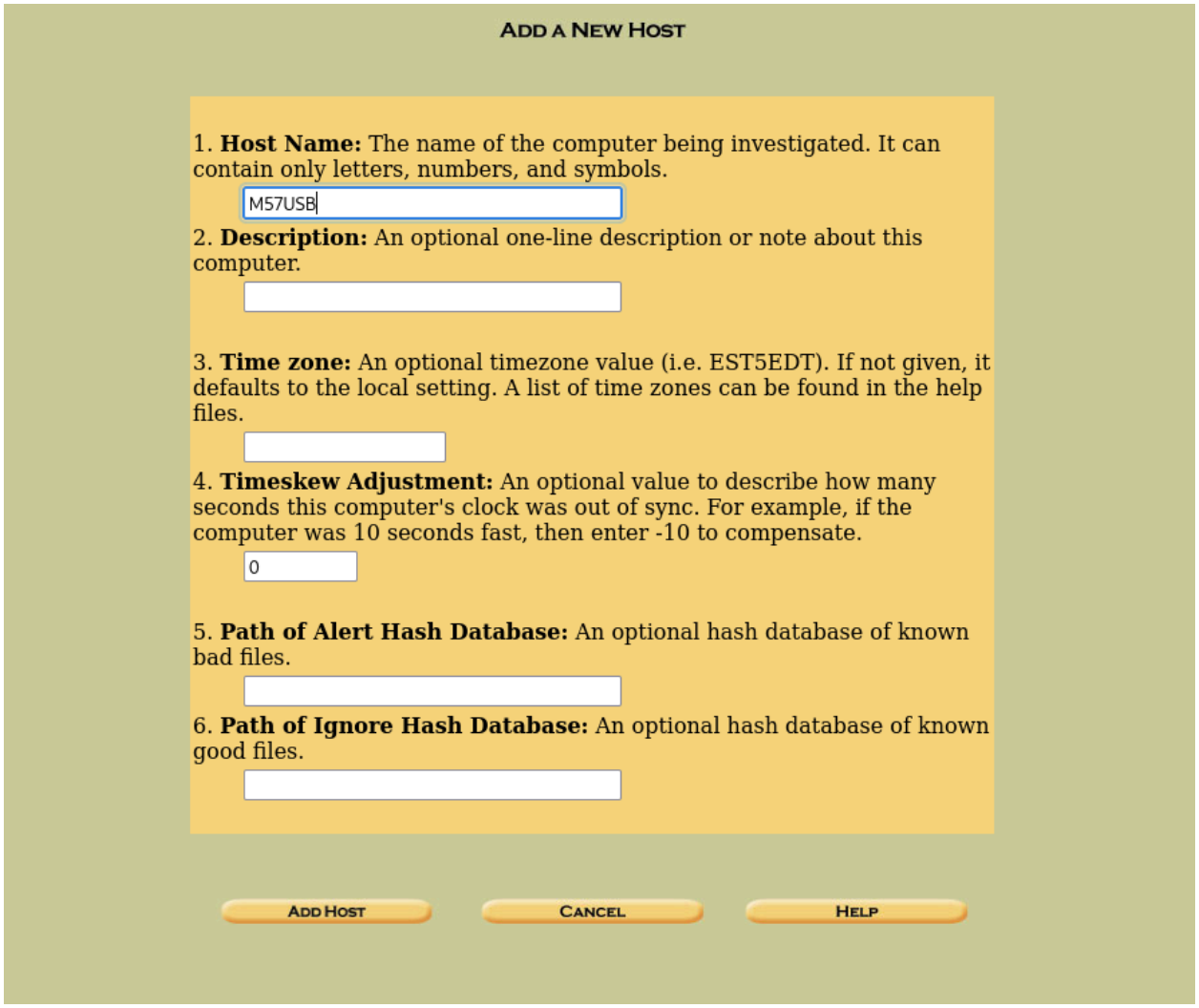
After creating the case I added a new host called M57USB. I left the other fields at their default values since they were not required.



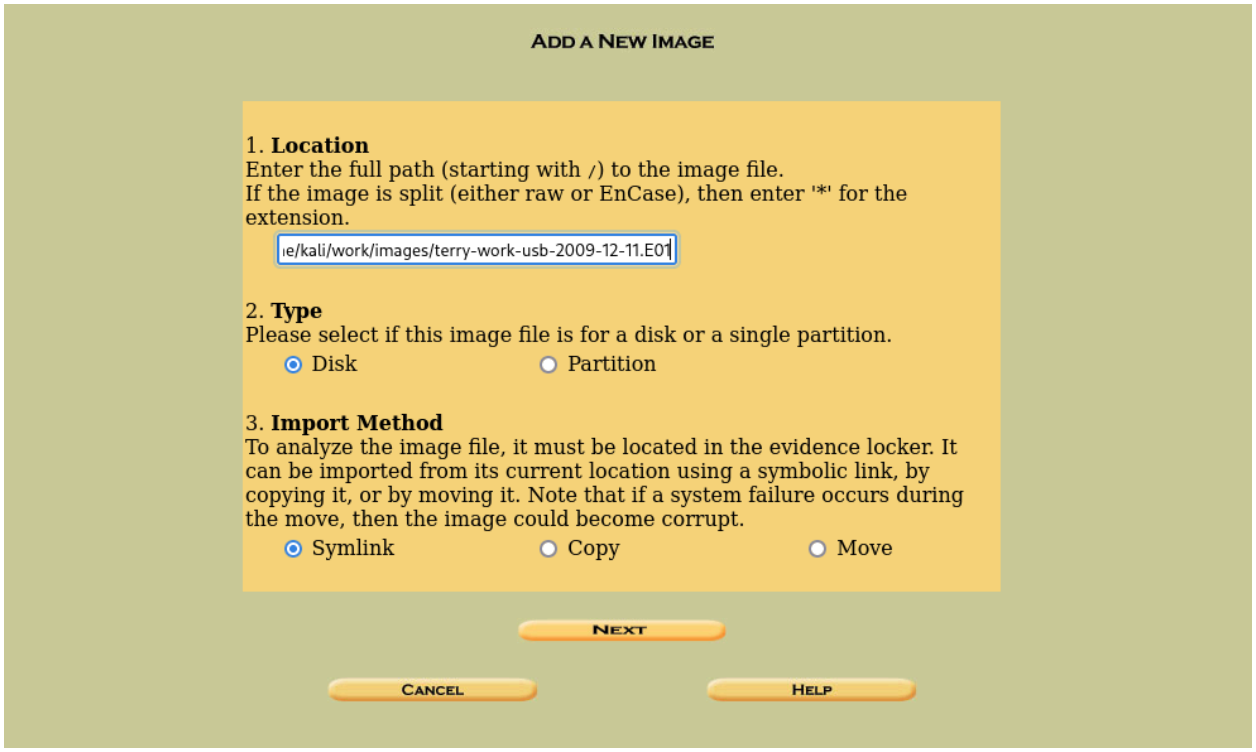
Next I added the forensic image terry work usb two thousand nine twelve eleven E01. I selected Disk as the type and used the symlink option for importing.



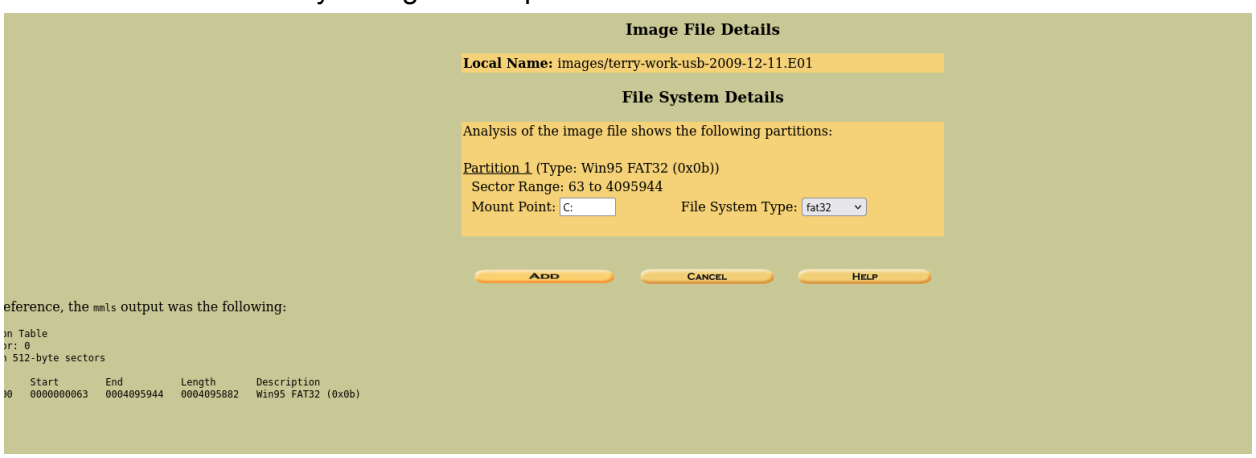
Autopsy showed that the image contained one main partition formatted as FAT32. The volume label was TERRYS WORK. Autopsy displayed the sector numbers and the file system layout.



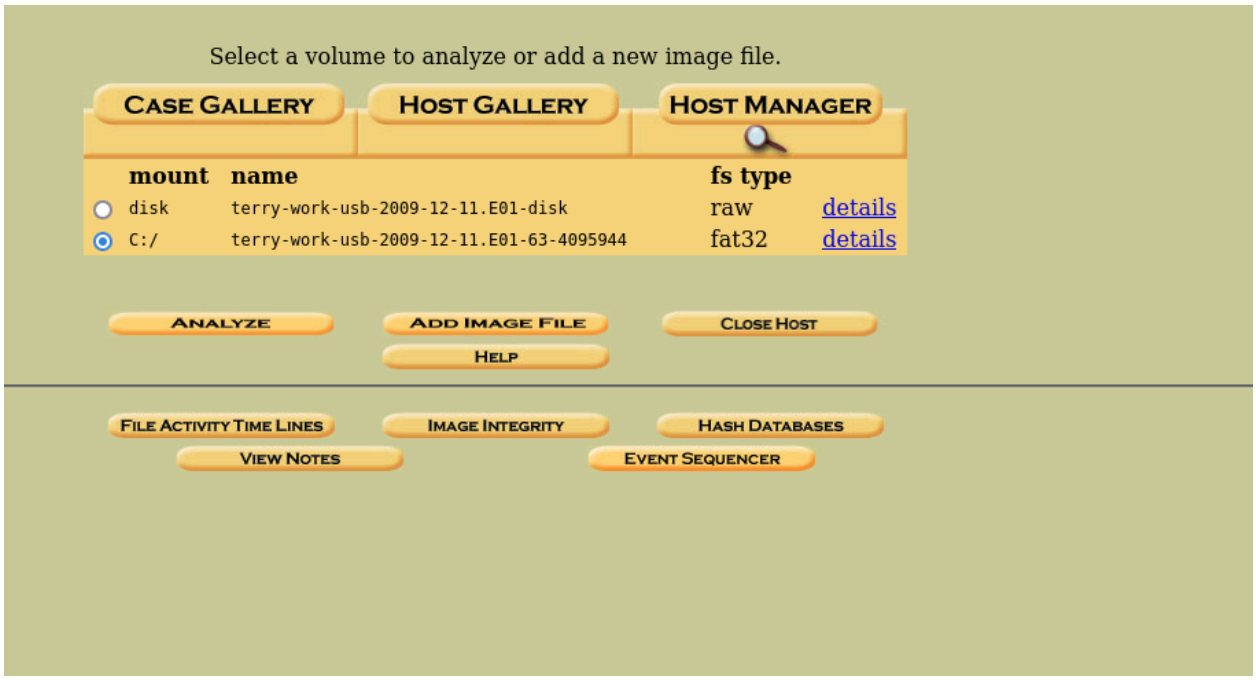
After loading the partition I opened the C directory. I reviewed the list of visible files including python scripts, text documents, images, and executable files.



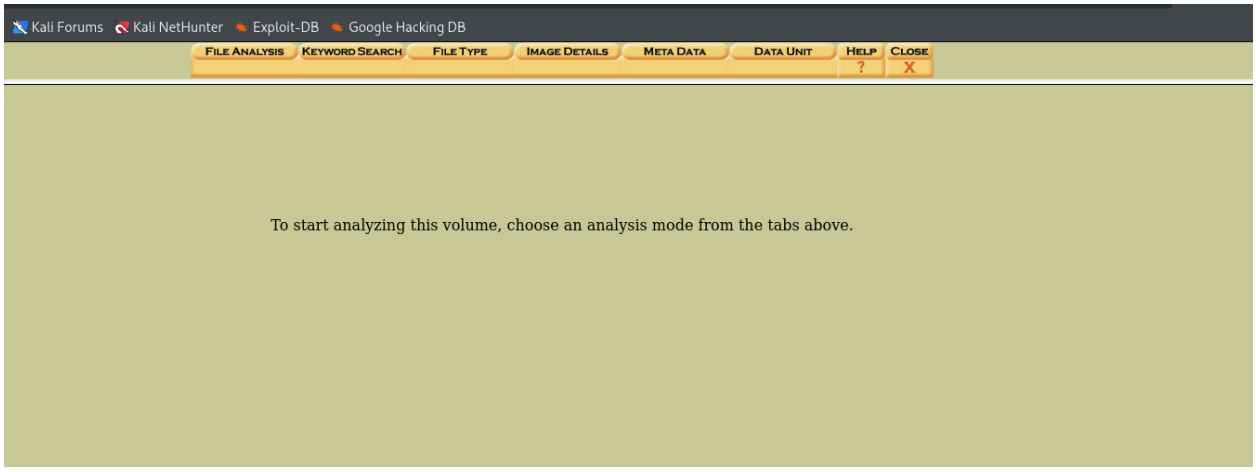
Autopsy generated MD5 values for all files in the directory. This helps confirm integrity and makes it easier to identify changes or duplicates.



I opened the metadata view for specific files. This showed MD5 and SHA1 values along with written, accessed, and created times. One example was the file named underscore tilde one dot TRA which had detailed timestamps and allocation information.



Autopsy displayed the layout of the FAT32 file system including cluster size, root directory location, reserved areas, and total sector count.



I opened the deleted files section and found multiple items including Trash folders, Spotlight indexing files, journal attributes, and other system related entries. These had timestamps from November seventeen two thousand nine and were linked to system activity.

General File System Details

FILE SYSTEM INFORMATION

File System Type: FAT32

OEM Name: BSD 4.4
Volume ID: 0x4a741208
Volume Label (Boot Sector): TERRYS WORK
Volume Label (Root Directory):
File System Type Label: FAT32
Next Free Sector (FS Info): 158074
Free Sector Count (FS Info): 3937808

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 4095881
* Reserved: 0 - 31
** Boot Sector: 0
** FS Info Sector: 1
** Backup Boot Sector: 6
* FAT 0: 32 - 4024
* FAT 1: 4025 - 8017
* Data Area: 8018 - 4095881
** Cluster Area: 8018 - 4095881
*** Root Directory: 8018 - 8025

METADATA INFORMATION

Range: 2 - 65405830
Root Directory: 2

CONTENT INFORMATION

Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 2 - 510984

FAT CONTENTS (in sectors)

I used the File Type Sorting feature to scan allocated and unallocated files. This feature sorts files by type and helps detect hidden or unusual content.

Sort Files by Type

View Sorted Files

File Type Sorting

In this mode, Autopsy will examine allocated and unallocated files
and sort them into categories and verify the extension.

This allows you to find a file based onits type and find "hidden" files.

WARNING: This can be a time intensive process.

I continued browsing through the root folder and sub folders to confirm the presence of all content. This included images, executables, python scripts, text files, and system folders.

localhost:9999/autopsy?mod=1&submod=2&case=Lab12_Ziyad&host=M57USB&inv=unknown&vol=vol2

OffSec | Kali Linux | Kali Tools | Kali Docs | Kali Forums | Kali NetHunter | Exploit-DB | Google Hacking DB

File Analysis | Keyword Search | File Type | Image Details | Meta Data | Data Unit | Help | Close

Directory Seek

Enter the name of a directory that you want to view.
C:/

View

File Name Search

Enter a Perl regular expression for the file names you want to find.

Search

All Deleted Files

Expand Directories

All Deleted Files

Type dir / in	Name	Written	Accessed	Created	Size	UID	GID	Meta
d / d	C:/.Trashes/_01	2009-11-17 13:34:58 (CST)	2009-11-17 00:00:00 (CST)	2009-11-17 13:34:58 (CST)	0	0	0	133
r / r	C:/.Trashes/._501	2009-11-17 13:34:58 (CST)	2009-11-17 00:00:00 (CST)	2009-11-17 13:34:58 (CST)	4096	0	0	135
d / d	C:/.fseventsd	2009-11-17 10:48:38 (CST)	2009-11-17 00:00:00 (CST)	2009-11-17 10:48:38 (CST)	0	0	0	10
r / r	C:/.Spotlight-V100/Store-V1/Stores/7680DE76-88D9-43B3-AC7E-3B02E7F38194/_-7.IND	2009-11-17 13:34:58 (CST)	2009-11-17 00:00:00 (CST)	2009-11-17 13:34:58 (CST)	4096	0	0	1433
r / r	C:/.Spotlight-V100/Store-V1/Stores/7680DE76-88D9-43B3-AC7E-3B02E7F38194/journalAttr.1	2009-11-17 13:34:58 (CST)	2009-11-17 00:00:00 (CST)	2009-11-17 13:34:59 (CST)	0	0	0	1483
r / r	C:/.Spotlight-V100/Store-V1/Stores/7680DE76-88D9-43B3-AC7E-3B02E7F38194/0.shadowIndexHead	2009-11-17 13:35:12 (CST)	2009-11-17 00:00:00 (CST)	2009-11-17 13:35:13 (CST)	4096	0	0	1493
r / r	C:/.Spotlight-V100/Store-V1/Stores/7680DE76-88D9-43B3-AC7E-3B02E7F38194/tmp.0.cmpt.indexPostings	2009-11-17 13:35:14 (CST)	2009-11-17 00:00:00 (CST)	2009-11-17 13:35:14 (CST)	182	0	0	1525
r / r	C:/.Spotlight-V100/Store-V1/Stores/7680DE76-88D9-43B3-AC7E-3B02E7F38194/tmp.0.cmpt.indexHead	2009-11-17 13:35:14 (CST)	2009-11-17 00:00:00 (CST)	2009-11-17 13:35:14 (CST)	4096	0	0	1528

File Browsing Mode

In this mode, you can view file and directory contents.

File contents will be shown in this window.
More file details can be found using the Metadata link at the end of the list (on the right).
You can also sort the files using the column headers

localhost:9999/autopsy?mod=1&submod=2&case=Lab12_Ziyad&host=M57USB&inv=unknown&vol=vol2

OffSec | Kali Linux | Kali Tools | Kali Docs | Kali Forums | Kali NetHunter | Exploit-DB | Google Hacking DB

File Analysis | Keyword Search | File Type | Image Details | Meta Data | Data Unit | Help | Close

Directory Seek

Enter the name of a directory that you want to view.
C:/

View

File Name Search

Enter a Perl regular expression for the file names you want to find.

Search

All Deleted Files

Hide Directories

C:/

Current Directory: C:/

Add Note | Generate MD5 List of Files

Del	Type dir / in	Name	Written	Accessed	Created	Size	UID	GID	Meta
Error Parsing File (Invalid Characters?): V/V 65405830: $OrphanFiles 0000-00-00 00:00:00 (UTC) 0000-00-00 00:00:00 (UTC) 0000-00-00 00:00:00 (UTC) 0000-00-00 00:00:00 (UTC) 0 0 0									
	v / v	$FAT1	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	2044416	0	0	65405828
	v / v	$FAT2	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	2044416	0	0	65405829
	v / v	$MBR	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	512	0	0	65405827
	r / r	.,,Trashes	2009-11-17 10:47:46 (CST)	2009-11-17 00:00:00 (CST)	2009-11-17 10:47:47 (CST)	4096	0	0	5
	r / r	.,M57biz.jpg	2009-11-17 10:49:24 (CST)	2009-11-17 00:00:00 (CST)	2009-11-17 10:49:25 (CST)	4096	0	0	17
	r / r	.,patentauto.py	2009-11-17 13:47:18 (CST)	2009-11-17 00:00:00 (CST)	2009-11-17 13:47:19 (CST)	4096	0	0	54
	r / r	.,patentterms.txt	2009-11-17 13:47:18 (CST)	2009-11-17 00:00:00 (CST)	2009-11-17 13:47:19 (CST)	4096	0	0	60
	r / r	.,urlscopyright.txt	2009-11-17 13:47:18 (CST)	2009-11-17 00:00:00 (CST)	2009-11-17 13:47:19 (CST)	4096	0	0	49
	r / r	.,urlscrypotography.txt	2009-11-17 13:47:18 (CST)	2009-11-17 00:00:00 (CST)	2009-11-17 13:47:19 (CST)	4096	0	0	43
	r / r	.,urlspatents.txt	2009-11-17 13:47:18 (CST)	2009-11-17 00:00:00 (CST)	2009-11-17 13:47:18 (CST)	4096	0	0	37
	r / r	.,urlspersona.txt	2009-11-17 13:47:18 (CST)	2009-11-17 00:00:00 (CST)	2009-11-17 13:47:18 (CST)	4096	0	0	31
	r / r	.,urlstime_machine.txt	2009-11-17 13:47:18 (CST)	2009-11-17 00:00:00 (CST)	2009-11-17 13:47:18 (CST)	4096	0	0	23

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Search for File Name

File Type:
AppleDouble encoded Macintosh file

MD5 of content:
5ecad39c470178e1b0ef93e534b60fda -

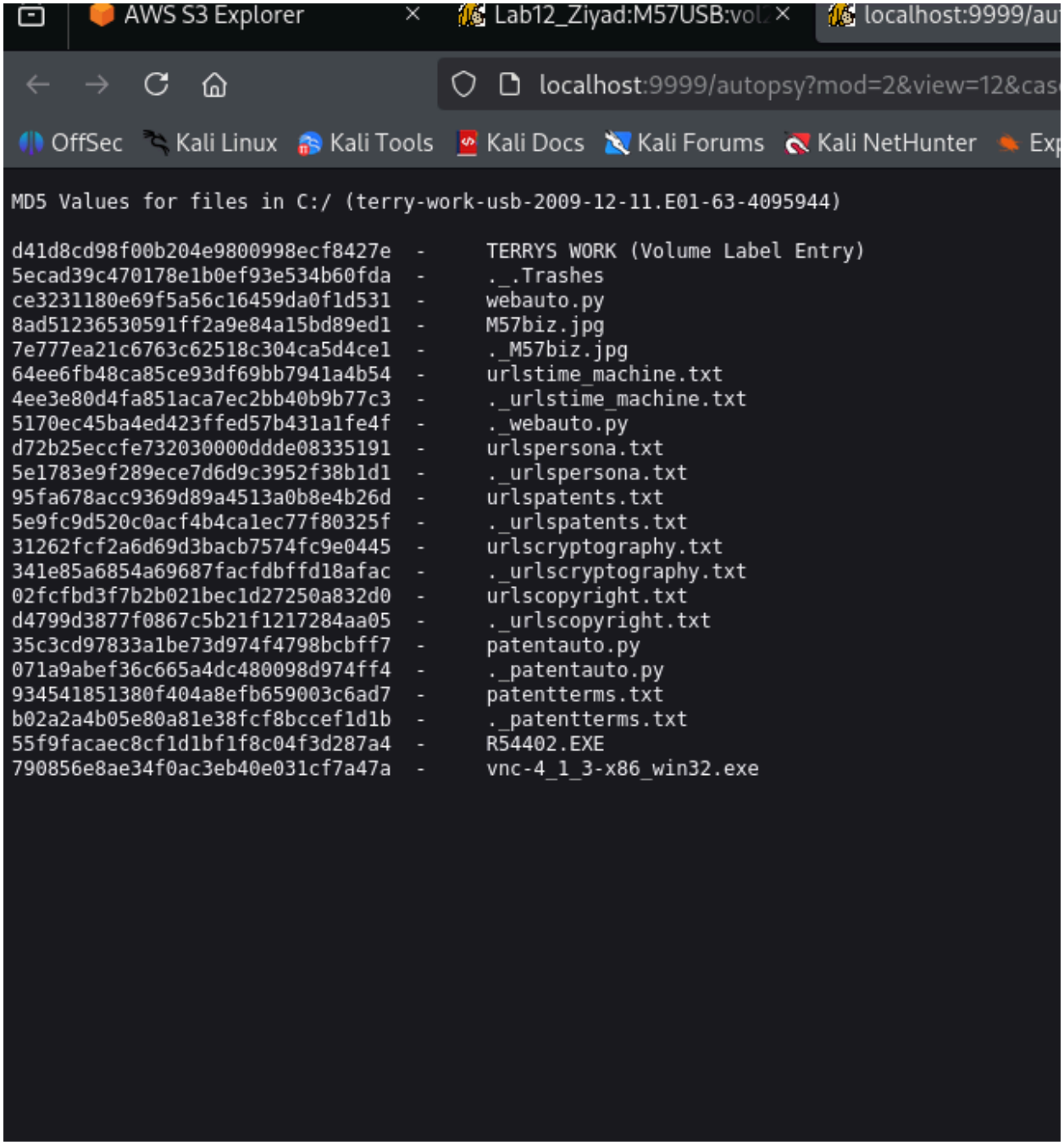
SHA-1 of content:
36dfed64b95c28cf63cce63da3d153cba020a88b -

Details:

Directory Entry: 5
Allocated
File Attributes: File, Hidden, Archive
Size: 4096
Name: _~1.TRA

Directory Entry Times:
Written: 2009-11-17 10:47:46 (CST)
Accessed: 2009-11-17 00:00:00 (CST)
Created: 2009-11-17 10:47:47 (CST)

Sectors:
8034 8035 8036 8037 8038 8039 8040 8041



This lab helped me learn how to use Autopsy to perform a full forensic analysis of a disk image. I created the case, added the host, loaded the image, explored the file system, checked file hashes, reviewed metadata, examined deleted files, and used type sorting. I documented every step with screenshots and gained a stronger understanding of how forensic tools handle evidence and present file system information.